Alles KLAR? A Rule-Based Simplicity Score for German Text Simplification

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Abstract

German law requires public authorities to provide written information in *Leichte Sprache* (LS, Easy Language), yet no automatic measure can confirm that a text offered as a simplification is any simpler compared to the original. Reference-based measures need scarce human-written simplifications to score against; reference-free readability formulas need none, but do not indicate *what* makes a text hard. We close this gap with KLAR, a reference-free simplicity metric for German text simplification. It is built from rules informed by the *LS* rulebooks and DIN SPEC 33429. Scores for each rule are normalized against a large unlabeled web corpus and aggregated into one number that shows which rules a text violates. Across four German corpora KLAR separates simplified from original text (AUC 0.65–0.95), and tracks human *simplicity* judgments while remaining independent of *meaning* judgments. Without any reference text it matches a learned reference-based metric at tracking simplicity. It also orders texts by difficulty levels assigned independently of the rules, falling across CEFR levels A1–C2 and rising across a register ordering from *Fachsprache* (FS, technical language) to *LS*. In practice, the rules behind KLAR drive an agentic simplification system that, with no task fine-tuning, matches the strongest of four fine-tuned German specialists on readability. Human raters prefer its outputs on meaning and faithfulness. We scope the validated claim to *simplicity* and treat meaning preservation as a secondary, reference-free check. The result is one validated, reference-free simplicity score, plus the rule-level trace of what to fix.

1 Introduction

German public authorities are legally required to provide written information in *Leichte Sprache* (LS, Easy Language)[1] so that official information stays accessible regardless of a reader's literacy or language proficiency. *LS* practice itself is codified in rulebooks (Maaß, 2015). Whether a given simplification is *good*, however, is not a single dimension but several, among them: (i) how simple the output is, (ii) how fluent it is, (iii) whether it preserves the original meaning, (iv) whether it fits the target group, and (v) whether it serves the communicative goal. We do not address other dimensions, such as acceptability, and organize the five above in Section 3.

Modern language models produce fluent simplification candidates, but no current system reliably satisfies all of these dimensions at once. Verifying a candidate needs aligned evaluation data, which German text simplification has for a limited set of corpora. Germany's federal accessibility-monitoring body reaches the same conclusion, finding that current AI tools "[...] do however not yet fully meet the requirements for comprehensibility and legally compliant accessibility. Current AI tools focus primarily on the linguistic level [...]; text structure and content are not currently translated in a manner tailored to the target audience."[2] (Überwachungsstelle des Bundes für Barrierefreiheit von Informationstechnik, 2025). The same body stresses that human expertise and review by a target-group panel (*Prüfgruppe*) remain essential. We take this requirement as our premise and aim to support such review.

Within that supporting role, a metric can contribute a first pass on the one dimension that is, by definition, a property of the text. That dimension is simplicity. *LS* is fixed by linguistic rules so its sim-

[1] Since 2018, § 11 of the Federal Disability Equality Act (*Behindertengleichstellungsgesetz*, BGG) obliges federal authorities to explain decisions, forms, and official documents in *LS* on request (Bundesministerium der Justiz, 2018).

[2] Original: *"[...] erfüllen jedoch die Anforderungen an Verständlichkeit und gesetzeskonforme Barrierefreiheit bisher nicht vollständig. Die aktuellen KI-Tools setzen hauptsächlich auf der Sprachebene an [...]; Textstruktur und inhaltliche Ebene werden nach aktuellem Stand nicht zielgruppengenau übersetzt."*

plicity is in principle readable from the text alone. Two families of tools already attempt this measurement. One family is reference-based (Xu et al., 2016) and relies on sentence- or document-aligned human references, which in German exist for only a handful of corpora, mostly news, and span few registers or target groups. The reference-free ones, classical readability formulas such as Flesch Reading Ease (Flesch, 1948), need no references. But they reduce a text to properties of sentences and words, they were never designed for *LS*, and they return a single number that does not say *what* made a text difficult. We therefore ask whether simplicity can be measured automatically and reference-free, well enough to stand in for the missing references and in a form a reviewer can act on.

It can. KLAR (*Kalibrierte Leichte-Sprache-Auswertung pro Regel*)[3] gives a single reference-free number that discriminates registers, tracks externally assigned ordinal levels, and, unlike a readability formula, shows which rules a text broke. Concretely, our contribution is threefold:

1. **A dimensional view of simplification** (Section 3). We separate what an evaluation can measure into properties of the input, output, and the input–output relation. This separation lets KLAR target one dimension, simplicity, in isolation from the others.

2. **KLAR, a reference-free simplicity metric** (Section 4), a combined score over calibrated rules informed by the *LS* rulebooks and DIN SPEC 33429. We validate it on four German corpora, by register discrimination (Section 5), agreement with expert human simplicity ratings, and concordance with externally assigned ordinal levels (Section 6).

3. **A system demonstration** (Section 7). We put KLAR's rules to work inside Klarbot, the agentic simplification system we built, and show it matches the strongest of four fine-tuned German specialists with no task fine-tuning, corroborated by a secondary reference-free meaning check (Section 7) and a direct human preference comparison (Section 7).

2 Related Work

Automatic text simplification rewrites a text to be easier to understand while keeping its meaning. Evaluating the simplification is an open problem. A metric has to decide whether a candidate is simpler, and for German text simplification it must do so against a regulated target. Evaluation methods can be divided into reference-based and learned metrics, which both need human reference simplifications, and reference-free, which do not. KLAR belongs to the last group. We review each in turn and place it among the German precedents our rules build on.

Reference-based metrics. Most simplification metrics are reference-based. SARI (Xu et al., 2016) scores add/keep/delete edit operations against human references, and BLEU (Papineni et al., 2002) (developed for machine translation) and BERTScore (Zhang et al., 2020) are also widely reported. For German, EASSE-DE (Stodden, 2024) bundles nine test sets with SARI, BLEU, BERTScore-Precision and German readability formulas, but every component needs parallel references and none of the test sets targets regulated *LS*. Alva-Manchego et al. (2021) show that such metrics correlate only moderately with human judgments and shift across systems and reference sets.

Learned metrics. These replace the fixed formula with a model trained to predict human quality judgments. LENS (Maddela et al., 2023) and its edit-level variant SALSA (Heineman et al., 2023) do this for English, and SLE (Cripwell et al., 2023) is a reference-free, *simplicity-only* learned metric. Such metrics need a corpus of human-rated simplifications to train on. The community benchmarks also stay English-centric, including the TSAR 2025 shared task on readability-controlled simplification (Alva-Manchego et al., 2025). For German, DEtect (Korobeynikova et al., 2026) adapts the LENS framework and is the closest competitor metric which is learned (trained on synthetic LLM scores), reference-based, and a black box. Ours is a different kind of metric, deterministic, explainable, and reference-free. We use DEtect itself as a comparator for KLAR (Section 5.5) and as an external check on our system (Section 7).

German readability and rule-based metrics. DeLite (vor der Brück et al., 2008; Glöckner et al., 2006) is the closest ancestor of our calibration. It puts heterogeneous readability indicators on a per-feature scale and combines them into a weighted sum, but it learns that aggregation from a human-rated corpus, so it is supervised. Klöser et al. (2026)

[3]The name is German for a calibrated, per-rule score of *LS*, and puns on *klar*, "clear".

build a reference-free, interpretable "fingerprint" of German simplifications, including a rule-based component over *LS* LanguageTool rules, but their goal is to tell simplification systems apart, not to produce a single validated simplicity score; we test convergence against it in Section 5.3. To our knowledge, no prior reference-free simplicity metric for German text simplification has been shown to track externally assigned ordinal levels.

Rule provenance and the regulatory frame. Our rules derive from the established *LS* rulebooks (Maaß, 2015; Bredel and Maaß, 2016; Netzwerk Leichte Sprache e.V., 2022), recently consolidated in DIN SPEC 33429 (Deutsches Institut für Normung, 2025); the broader *Einfache Sprache* (ES, Plain Language) was separately standardized in DIN 8581-1 (Deutsches Institut für Normung, 2024). These rulebooks are prescriptive, and their validity has been questioned (Bock, 2019). We therefore treat our rules as informed by these rulebooks and the DIN SPEC, not as ground truth. Reader proficiency is a separate axis. The CEFR scale (A1-C2) defines reader levels (Council of Europe Education Department, 2024), and CEFR grading for German has been modeled as a learner-proficiency task (Hancke and Meurers, 2013). The BFIT-Bund statement (Überwachungsstelle des Bundes für Barrierefreiheit von Informationstechnik, 2025) frames the gap our work addresses.

Evaluation data and systems. We draw on six German corpora, grouped by how we use them (full dimensions in Appendix A, Table 2). Four are *contrast* corpora that differ in how far the simplification moves from its original: APA-LHA (Spring et al., 2021) and the Simple German corpus of Toborek et al. (2023) are high-contrast. The web and APA-news subsets of DEplain (Stodden et al., 2023) are low-contrast, with the APA news the hardest case and disjoint from the rest. SimpEvalDE (Korobeynikova et al., 2026) is a ratings corpus, adding expert *simplicity* and *meaning* scores on 360 system outputs over 60 originals. CEFR-graded `ELG-CEFR-DE` (Breuker, 2023; Imperial et al., 2025) is not a contrast corpus but an independent ordinal scale, used in Section 6 to test graded ordering. Toborek mixes both German controlled languages, *ES* and the stricter *LS* (Section 3 separates the two); the other corpora are *ES*. For the system demonstration (Section 7) we compare our agentic system against four fine-tuned German specialists: erlesen-leo-7b and -13b (Klöser et al., 2024), German4all (Anschütz et al., 2025), and ELGEPA-8b (Schomacker et al., 2026).

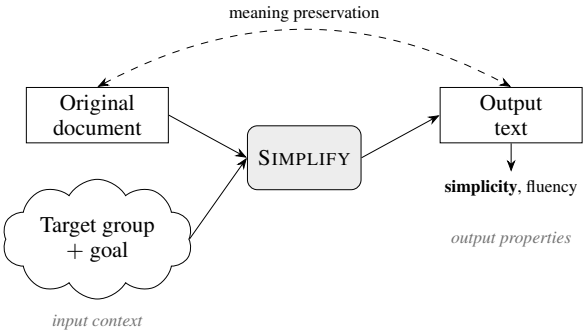


Figure 1: Dimensions of simplification evaluation. The *input context* (target group, goal) configures the task; the mechanism produces an output whose *output properties*, **simplicity** (the validated path) and fluency, are measurable from the output alone; *meaning preservation* is an input–output relation. This work validates simplicity, checks meaning preservation as a secondary axis, and scopes the rest.

3 A Dimensional View of Simplification Evaluation

Discussions of simplification quality often conflate distinct things. Here we fix a task definition and a small set of evaluation dimensions (Figure 1).

Given an original document and a *context*, a simplification mechanism produces an output text. An evaluation can then measure:

- **Properties of the output:** *simplicity* (how linguistically demanding the text is) and *fluency* (whether it is well-formed German).
- **An input–output relation:** *meaning preservation*, whether the output keeps the original content (*meaning*) without adding falsehoods (*faithfulness*).
- **Properties of the input context:** the *target group* (who the text is for) and the *goal* (to inform, to instruct, to give legal notice).

The literature often treats two notions interchangeably, a property of the output and a property of the target group (the intended reader's proficiency). *Leichte Sprache* (LS) and *Einfache Sprache* (ES) are properties of the text, German *controlled languages*.[4] They occupy the accessible

[4]Following Hennig (2022), *Leichte Sprache* is not a language variety or register in the sociolinguistic sense. An organically emergent variety forms through convergent usage

end of a continuum that extends upward through *Standardsprache* (StS, standard language) to *Fachsprache* (FS, technical language). LS is openly codified in published rulebooks and mandates review by a target-group panel. ES is the broader, less restricted form. It is governed by DIN 8581-1 (Deutsches Institut für Normung, 2024), which sits behind a paywall, so its precise requirements are not publicly available. CEFR levels (A1–C2), by contrast, describe the *proficiency of the intended reader*, not a property of the text (Breuker, 2023). A simplification aims to bring a text's demand below a target reader's proficiency; register and proficiency are therefore distinct axes, and we keep them separate throughout.

To our knowledge, no existing metric covers these dimensions *jointly* and reference-free. This paper contributes a validated metric for the simplicity dimension (Section 4–Section 6) and puts its rules to work in a system we built (Section 7). We evaluate that system on two dimensions of our framework, simplicity and meaning preservation, the latter through a reference-free meaning check we also contribute (Section 7). Fluency, target group and goal are discussed as scope boundaries (Section 9).

4 The Metric

KLAR is a single combined score over 20 rules informed by the *LS* rulebooks and DIN SPEC 33429 (e.g. sentence splitting, short common words, avoidance of subordinate clauses; the full definitions are in Appendix B). Each rule returns a raw score on its own scale. KLAR turns these scores into one calibrated, reference-free number.

KLAR's 20 rules cover only the part of the *LS* rulebooks a deterministic check can measure. Of the 47 rules in the Netzwerk Leichte Sprache rulebook (Netzwerk Leichte Sprache e.V., 2022), 17 govern document layout and a further step mandates target-group review; the remaining 30 cover words, numbers, sentences, and text. KLAR's rules come from this text-level part, and guidelines that need human judgment fall outside it. It does not, for example, enforce third-person-pronoun avoidance.

in a speaker community. *Leichte Sprache* is instead fixed by an explicit top-down rule set. We treat it as a *controlled language* (Madina et al., 2023), a rule-governed, reductive mode of Standard German.

Per-rule calibration. The raw rule scores are not comparable. Each rule measures a different feature, so each has its own range and its own baseline in everyday German. We estimate that baseline, *typical German*, from the rule's distributional average over a large web corpus. Some rules score near 1 on almost any text; others vary widely (Appendix Figure 5a). To combine them, we *calibrate* each rule in two steps. First we *normalize* it by estimating the rule's typical level on a large *unlabeled* German web corpus, the German split of C4/mC4 (Raffel et al., 2020; Xue et al., 2021)[5], and z-score the raw value against that baseline. Then we pass the z-score through a sigmoid into a *calibrated score* in $[0, 1]$:

$$\mathrm{cal} = \sigma(k\, z), \quad z = \frac{x - \mu}{\sigma}, \quad k = 2. \tag{1}$$

Every rule now reads the same way. A calibrated score of 0.5 is typical German, above 0.5 is simpler than typical, below is harder. We borrow this calibration from the supervised DeLite checker (vor der Brück et al., 2008), but anchor it on an unlabeled corpus with no human ratings and fix the slope $k = 2$ rather than tune it. Each rule is directional by construction, measuring how far a text moves toward one LS target such as shorter sentences or more everyday words. Accordingly, the web corpus fixes only the scale while the direction of improvement comes from the LS rulebook.

Reliability weighting. A rule backed by little evidence in a document should count for less. We weight each rule by:

$$w_i = \frac{n_i}{n_i + n_{0,i}} \tag{2}$$

where n counts how often the rule applies in the document[6], and n_0 is the support at which the rule earns half weight (Figure 5b). A rule with little support is discounted; a well-supported one counts in full. Each rule's n_0 is read off the C4 corpus (roughly 15 to 160 across rules). The form $n/(n + n_0)$ is the credibility shrinkage factor (Bühlmann, 1967; Kim and Zhang, 2014); we use only this factor, not the full model with its prior, so we call it a reliability weight.

KLAR is the reliability-weighted mean of the calibrated rule scores, one number in $[0, 1]$ that

[5] We sample 30k documents of $\geq$ 100 words from the `allenai/c4` release.

[6] Some rules are scored per sentence, others per token.

needs no reference text. Because it is only a weighted mean, it also decomposes: each rule's calibrated score can be read on its own, so a low total points to the specific rules a text broke.

Readability indices. We also compute three classical indices: the German Flesch Reading Ease (FRE) (Flesch, 1948; Amstad, 1978), the Wiener Sachtextformel (WSF) (Bamberger and Vanecek, 1984), and LIX (Björnsson, 1968). These are not part of KLAR; we report them separately as external, independent points of comparison.

Scope. KLAR is defined for texts of $\geq$ 100 words (the MW100 scope used throughout); Appendix B.2 states the assumptions in full.

5 Validation: Convergent but Not Redundant with Readability Indices

We validate KLAR on static, human-written texts with four checks. KLAR must tell simplified text from its original (Section 5.1). It should agree with the established readability indices while still measuring something they miss (Section 5.3). Each rule's contribution should read on its own, so the score says *why* a text is difficult (Section 5.4). KLAR should agree with expert human ratings (Section 5.5). Whether KLAR also orders texts by externally assigned difficulty is taken up separately in Section 6.

5.1 Discrimination

A minimal requirement is that KLAR scores known-simplified text above its original. KLAR does so on all four corpora (Table 4), with AUC 0.65 to 0.95[7]. The AUC tracks each corpus's contrast (either high or low), highest on APA-LHA and Toborek and lowest on the DEplain subsets, with DEplain-APA the hardest. On Toborek, which carries both controlled languages, the rules separate simplified text from its original in each (*ES* AUC 0.98, $n = 168$; *LS* 0.80, $n = 201$). These subsets also differ in original domain, so the AUC gap between them reflects domain, not register.

5.2 The normalization tradeoff

Normalization makes the rules comparable. Each rule then reads as how far a text has moved toward that rule's target relative to typical German, and the total decomposes into named rules. Does this explainability cost accuracy? On high-contrast Toborek the raw combined score, before normalization and weighting, discriminates better than KLAR (AUC 0.986 vs. 0.895; the drop is significant, $\Delta = -0.091$, 95% confidence interval (CI) $[-0.203, -0.031]$). On low-contrast DEplain-web the three combined scores are equivalent (Table 3 decomposes the drop into its normalization and weighting steps). The cost is small and confined to the easy case, and only the normalized scale lets KLAR say *why* a text is difficult.

5.3 Comparison with the readability indices

KLAR and the classical readability indices both target text simplicity, so they should correlate (Table 5).[8] The agreement is moderate in high-contrast scenarios (Spearman ρ near 0.6 on Toborek) and weak for low-contrast ($\rho = 0.15$–0.2 on the two low-contrast DEplain corpora). KLAR and the indices therefore measure related but not identical features. A per-rule breakdown shows the agreement comes from a single shared length feature, not the rule set as a whole (Appendix D).

No single signal wins everywhere at separating simplified text from its original (Table 8). The length-based indices discriminate best where simplification is largely length reduction, on Toborek and the DEplain-APA news, while KLAR separates best on DEplain-web, where the simplification is not mainly about length.

The closest prior metric is the Simplification Profiler of Klöser et al. (2026), the only other reference-free and explainable German metric. It shares KLAR's LanguageTool rulebook, so the two agree on simplicity by construction but diverge on meaning, which makes KLAR distinct rather than redundant. As the Profiler's authors release no code, we re-implement it from the paper's prose (Appendix E).

5.4 Per-rule decomposition and steering

Because KLAR is a weighted mean, each rule's calibrated score reads on its own, so a low total points to the specific rules a text broke. On Toborek, 17 of the 20 rules separate simplified text from its original (Benjamini–Hochberg false-discovery rate[9], BH-FDR < 0.05), and the signal concentrates in

[7] AUC here is the chance that a randomly drawn simplified text outscores its own original, so 0.5 is a coin flip and 1 is perfect separation

[8] APA-LHA is graded, so we examine it under grading in Section 6 rather than here.

[9] The Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995) controls the false-discovery rate, the expected share of false positives among the rules called significant, across the 20 simultaneous tests.

a few structural rules (Table 6). Sentence splitting and short words carry the most; reader address and active voice add more; question avoidance runs the other way, because simplified texts ask the reader more direct questions than their originals, often as headings.

Which rules fire depends on the domain. Sentence splitting and short words separate simplified text from its original on all four corpora and form a stable backbone. The remaining rules shift by domain, so those that flag administrative news differ from those that flag Toborek's encyclopedic texts. We therefore read the per-rule scores as a diagnosis of what makes a given text hard.

Because the rules are separable, KLAR can be *steered*. Disabling or reweighting a rule moves the score toward a chosen profile, which a single readability number cannot do.

5.5 Agreement with human ratings

The strongest evidence comes from human judgment. On SimpEvalDE, KLAR tracks the mean human *simplicity* rating ($n = 360$; $\rho = 0.31$, 95% CI $[0.17, 0.44]$, $p < 10^{-8}$) while remaining independent of the *meaning* rating ($\rho = 0.03$, CI spans zero). A simplicity metric should not also move with meaning, so this independence shows KLAR isolates simplicity; meaning is taken up by the separate check of Section 7. These texts are single sentences, below the 100-word scope, so the reliability weight pulls each score toward the typical-German prior and attenuates the correlation; $\rho = 0.31$ likely understates the agreement on in-scope documents.

KLAR reaches this agreement without using any reference text. On the same pairs it is statistically on par with DEtect (paired $\Delta\rho = +0.03$ in DEtect's favor, cluster-bootstrap 95% CI $[-0.15, +0.21]$; the marginal ρ are 0.31 for KLAR and 0.35 for DEtect). This holds even though DEtect scores each text against a human reference simplification that KLAR never uses, and SimpEvalDE is DEtect's own validation set. Matching a learned reference-based metric on its home ground, with no reference and no training of our own, is evidence that KLAR measures what raters call simplicity.

Figure 2 places KLAR against the competing metrics on the two properties that distinguish them, reference-free and explainable (full numbers in Appendix E).

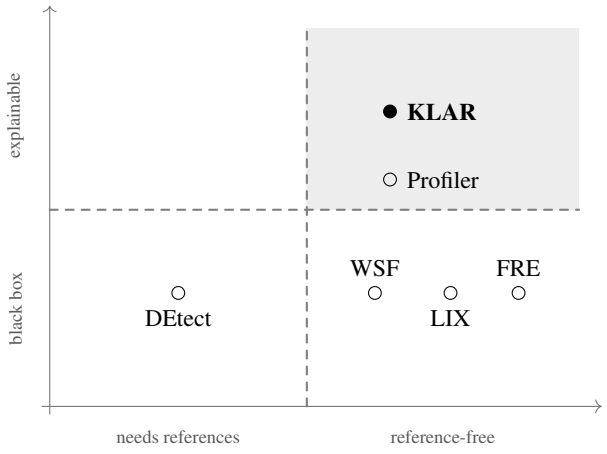


Figure 2: Simplicity metrics by two properties: reference-free (x) and explainable (y). Only KLAR and the Profiler occupy the reference-free, explainable quadrant (shaded). KLAR (filled star) is a single validated simplicity score; the Profiler (open star) is a multi-axis system fingerprint. The readability formulas are reference-free but not explainable; DEtect is learned and reference-based. Discrimination AUC and human-simplicity ρ for every metric are in Table 8.

6 Ordinal Grading

Section 5 showed that KLAR tells a simplified text from its original. A harder test is whether the score tracks *how far* a text has been simplified, ordering texts by a scale set independently of the metric. Passing this test would show that KLAR measures graded difficulty, not just a binary simplified-or-not contrast. We test it against three kinds of externally assigned ordinal levels: CEFR levels, the graded levels of a simplification corpus, and a register order. These labels are coarse, and experts agree only loosely on readability (Collins-Thompson, 2014), so we read agreement as concordance with an external ordering, not as a recovered ground truth.

CEFR levels. On the CEFR-graded `ELG-CEFR-DE` corpus (MW100 scope, $n = 493$ of 509; A1–C2), KLAR falls monotonically as texts get harder (Figure 3; Spearman $\rho = -0.50$; the full corpus including shorter texts gives $\rho = -0.51$). The ordering is sharpest across the easy half, where adjacent bands separate, and saturates at the hard end, where the B2–C2 bands are no longer distinguishable. As with discrimination (Section 5.2), calibration costs ordinal rank. The raw combined score grades CEFR at $\rho = -0.71$ (paired $\Delta\rho = -0.21$, 95% CI $[-0.26, -0.16]$), and the readability indices grade at least as well ($\rho \approx 0.74$–0.76). KLAR adds a reference-free,

decomposable ordering, not a sharper one.

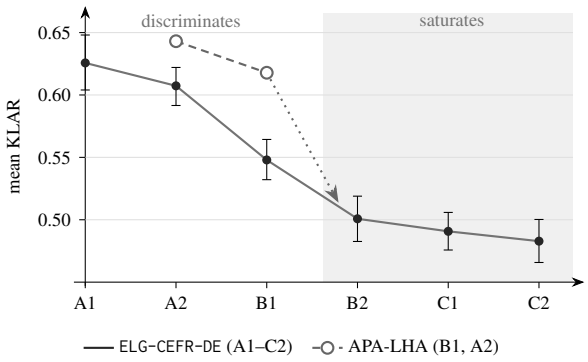


Figure 3: KLAR across CEFR levels (band means, 95% bootstrap CI; MW100, $n = 493$). The score falls monotonically from A1 to C2 ($\rho = -0.50$), separating adjacent bands on the easy half and saturating at B2–C2 (shaded). Open markers overlay APA-LHA's B1 and A2 targets (CIs narrower than the markers); the dotted tail points toward the off-scale original ($\rho = 0.62$, see text).

Graded simplification (APA-LHA). On the 1344 APA-LHA news triples (original, B1, A2),[10] KLAR rises with each simplification step (Spearman $\rho = 0.62$); B1 outscores the original in 90% of triples and the full *original* < *B1* < *A2* ordering holds in 57%.

Register placement. KLAR also orders texts by *register*, not only by the gap between a simplification and its original. It places four German registers along the technical-to-plain ladder *FS* < *StS* < ES < LS, where *FS* is expert-facing technical language (Spearman $\rho = 0.71$; Table 1). FS, StS, and ES separate at non-overlapping intervals, so the steps down from technical to simplified language are clean. Only the last step is a close call. The two controlled languages, ES and LS, sit in the right order, but on human text the gap CI crosses zero (Table 14). That step does separate on machine-generated text. German4all is itself a machine simplification system that emits both controlled languages. On its two output levels ($n = 514$), KLAR places *original* < ES < LS with every step CI-separated, and the ES-to-LS gap that crossed zero on human text separates here ($+0.024$, 95% CI $[0.010, 0.038]$; Table 14). A per-rule view shows the difference is lexical, not one of overall simplicity. LS adheres more closely than ES to the simple-word and sentence-fragment rules.

[10]These 1344 triples are the full APA-LHA set, not the smaller MW100 pair subset used for discrimination in Table 4, which is why the count differs.

Register	n	mean KLAR [95% CI]
FS	352	0.407 [0.394, 0.420]
StS	749	0.480 [0.453, 0.519]
ES	534	0.595 [0.568, 0.616]
LS	215	0.619 [0.597, 0.651]

Table 1: Register placement (higher = simpler). The ES–LS gap is the one contrast whose CI crosses zero (see text).

7 System Demonstration

We built Klarbot, an agentic simplification system that pairs a Translator/Supervisor loop with a question–answer comprehension check, where the Supervisor steers each rewrite with KLAR (Section 4). It runs on a pinned open-weights base model (Qwen3.5-122B-A10B) with *no task fine-tuning*. We evaluate its output against the four fine-tuned German specialists of Section 2 on the same documents, a paired pool of $n = 997$ from Toborek, DEplain-web, and DEplain-APA. We compare them three ways: readability, a meaning-preservation check, and overlap with human reference simplifications. The first two need no reference text and cover two dimensions of our framework (Section 3), simplicity and meaning preservation; the overlap measure needs the references. Figure 4 summarizes the result. Klarbot is the only system that stays in the top band on every axis.

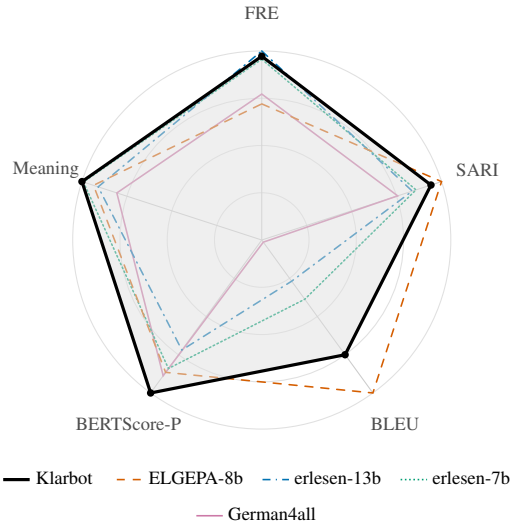


Figure 4: Five evaluation axes, normalized per axis (outer ring = best system, higher is better). Klarbot is the only system in the top band on every axis. Normalization, pool sizes, and exact values: Appendix F.

Reference-free readability. On the independent FRE score (not part of KLAR), Klarbot is statistically *equivalent* to both fine-tuned *erlesen* specialists (Section 2) despite using no fine-tuning (TOST, ± 5 FRE margin, 90% CI inside the band), and scores well above German4all and ELGEPA-8b (paired-document $\Delta = +17.8$ and $+22.0$).[11]

Reference-based ranking. Where references exist we also run the standard EASSE-DE harness (Stodden, 2024) (full values in Table 9, Appendix F). Klarbot and ELGEPA-8b sit together at the top. ELGEPA holds the SARI and BLEU point estimates and Klarbot the BERTScore-Precision one, but the Klarbot–ELGEPA gap is not significant on SARI (cluster-bootstrap $\Delta = -3.1$, CI $[-5.8, +7.3]$) or on BERTScore-P. Klarbot significantly outranks the other three specialists on SARI and both *erlesen* models on BERTScore-P; the only gap favoring a specialist is ELGEPA's lead on BLEU.

Meaning preservation. A simplicity score is silent on meaning (Cripwell et al., 2023), so a system could raise it by dropping content. We therefore pair it with a reference-free meaning check based on natural-language inference. Klarbot has the lowest content-deletion rate of the five systems under mDeBERTa (0.644) and ties the best under XLM-R (per-system values under both NLI models in Table 10), and it sits mid-cluster on the learned, reference-based DEtect metric (Korobeynikova et al., 2026).

Human preference. Human judgment is the direct test, but raters are expensive, so we tested only the hard cases. A panel of LLM judges voted on every Klarbot-versus-specialist pair, and we kept the ones it was split on, where the two systems were closest. Three trained linguists then judged these pairs on the two sides of meaning preservation (Section 3), *meaning* (whether the original content is kept) and *faithfulness* (whether anything false is added). They preferred Klarbot and almost never a specialist, by 99 to 14 on meaning and 69 to 13 on faithfulness (rest ties). Inter-annotator agreement and the panel–human agreement ceiling are reported in Table 12. Because these are the hardest pairs, the counts are a conservative floor, and the automatic axes point the same way. The sampling, the agreement ceiling, and the panel's limits are in Appendix G and Section 9.

8 Conclusion

Simplicity can be measured automatically, with no reference text to compare against. KLAR combines rules informed by the *Leichte Sprache* rulebooks and DIN SPEC 33429 into one calibrated number that, unlike a readability formula, shows which rules a text broke. That number separates simplified from original text across four German corpora, tracks expert simplicity ratings, and orders texts by CEFR and register levels assigned independently of the rules. The same rules also drive Klarbot, an agentic system that, with no task fine-tuning, matches the strongest of four fine-tuned German specialists on readability and ties it on reference-based SARI. Human raters, judging the closest system-versus-specialist pairs, prefer the system's outputs on meaning and faithfulness. Because the system carries no fine-tuning of its own, a stronger German base model can drop into the same pipeline at no retraining cost. Running that pipeline on a smaller same-family base (Qwen3.5-35B-A3B) reproduces the readability result, which model size alone does not explain.[12]

Most German registers and target groups are unlikely to ever have aligned gold. A transparent, rule-level score lets authors measure and iterate exactly where a reference-based metric cannot run. KLAR does not replace the human target-group panel; it hands the panel a measurable first draft to finalize.

9 Limitations

Simplicity only. The validated claim covers simplicity alone. Meaning preservation gets only a secondary, reference-free check (Section 7), not a validated claim of its own. Fluency, target group, and goal (Section 3) are out of scope, and target-group adaptation by reweighting rules per audience is a natural extension we do not test.

ES vs. LS and scope. KLAR does not separate the two controlled languages on human-written text, though it does on machine-generated output (Section 6); it is defined only for ≥ 100-word texts, and saturates among already-complex texts at the

[11] ELGEPA's published code and paper leave decoding unspecified. Under greedy decoding 26% of its outputs degenerate into repetition loops; with repetition blocking the gap narrows to $\Delta = +7.7$ and Klarbot still leads ($p = 1.8 \times 10^{-12}$).

[12] The 35B base is a mixture-of-experts with about 3B active parameters against the specialists' dense 7–13B, a within-family scale-down rather than a parameter match.

hardest CEFR levels. The rules and their resources are also German-specific. The calibration pipeline is language-agnostic, but porting KLAR to another language means rebuilding the rulebook and word lists, which we have not attempted.

Human evaluation of our own outputs. The human study behind the preference result (Section 7) is small, testing only the closest, hardest-to-call pairs, so it is suggestive rather than conclusive. It establishes what is, to our knowledge, the first human–human agreement ceiling for German simplification evaluation. That ceiling is low, with even trained linguists agreeing only weakly. Against this baseline, the LLM-judge panel tracks human *meaning* ratings about as closely as two raters track each other, though it remains a weak proxy for *faithfulness*. Consequently, we never treat the LLM as a validated replacement for human judgment. Separately, the combined-score *level* tracks human standalone-quality ratings no better than a readability formula, and their meaning ratings not at all. The score therefore reads simplicity, not meaning (Section 5.5). Full numbers are in Appendix G; a fully powered evaluation of Klarbot's outputs remains future work.

Reproducibility. The rule implementation and the generation system are a closed product. We release the formulas, operational rule definitions, and per-document score tables,[13] so the reported statistics are auditable. Exact rule scores are implementation-dependent, so the specification lets others re-implement and approximate our numbers rather than bit-reproduce them.

Lay Summary

Many people in Germany need information they can read with ease. The law says public bodies must offer this information in Easy Language, a form of German with short sentences and common words. Someone still has to judge whether a text really is simpler than the one it came from, and today people do this by hand. Some tools compare a text against a human-written example, but good examples are scarce and expensive. Other readability tools return a single number and stop there. Can we judge how simple a German text is from the text alone, and show a writer what to fix?

We built KLAR, a way to measure how simple a German text is using the text alone. KLAR checks a text against rules taken from the official Easy Language guidelines, such as short sentences and everyday words. It turns these checks into a single score and points to the rules a text should follow more closely, so a writer sees exactly what to improve.

We tested KLAR on four collections of German texts. It reliably separated simplified texts from their original versions, and it placed texts in the right order of difficulty, from beginner to advanced. Its scores matched how human readers judged simplicity. It also stayed independent of whether a text kept its original meaning, which we measure with a separate check.

We also used the rules behind KLAR to guide an automatic simplification system. Even as a general-purpose system, it matched the best of four specialized German systems on readability, and human readers preferred its results for keeping meaning and staying faithful to the source.

In short, KLAR offers a single, reliable score for how simple a German text is, together with a clear list of what to improve. It covers only the simplicity part of the task, so a person still needs to check that a text keeps its meaning and fits its readers, and KLAR is there to support that review. We hope it helps writers, public bodies, and researchers make information easier to understand for everyone.

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A Datasets

Table 2 collects the corpora used in this paper: the six evaluation corpora introduced in Section 2 and German4all, whose machine-generated outputs we also use as a register-placement dataset (Section 6). The four contrast corpora differ in how far the simplification moves from its original (high or low contrast); the remaining corpora supply ordinal levels or human ratings rather than a simplified-original contrast. A dash marks a dimension the corpus does not carry. Counts are the $\geq$ 100-word (MW100) figures used in the corresponding analysis; APA-LHA is 208 MW100 pairs for discrimination and 1344 triples for grading (Section 6).

B Operational Rule Definitions

Section 4 defines KLAR through its calibration and weighting. This section supplies the rule-level detail behind that definition, giving the operational form of each of the 20 rules, the scoring convention they share, and the readability indices reported next to KLAR but never folded into it. Every rule returns an *adherence* score in $[0, 1]$, where 1 marks full adherence to the simplification guideline (simpler) and 0 a full violation (more complex), so a higher raw score always means simpler. KLAR itself is the reliability-weighted mean of the 20 calibrated rules listed in Section B.1; the three readability indices (Section B.3) are reported separately and are not part of it.

B.1 The 20 rules in KLAR

Implementation. Every rule reads from a single spaCy 3.8.2 parse of the document with the `de_core_news_md` German model, which supplies the tokens, part-of-speech tags, dependency labels, and sentence boundaries the rules use (sentences come from spaCy's segmenter, `doc.sents`); *words* are alphabetic tokens (`token.is_alpha`). Two length rules score against fixed baselines taken from the German news subcorpus (`deu_news_2023`) of the Leipzig Corpora Collection (Goldhahn et al., 2012): a mean sentence length of 15.39 words (`Keep Sentences Short`) and a word-length normalizer of 13.89 (`Use Short Words`). Both are corpus statistics, not values tuned on evaluation data, which is what keeps the thresholds independent of our test sets. The lexical rules draw on three resources: a 4,616-word everyday-German list, the top $\sim$1% by frequency of the Leipzig German web corpus (`deu-com_web_2021`) (Goldhahn et al., 2012); a 2,706-entry idiom list from the German Wiktionary directory of idioms (*Verzeichnis: Deutsch/ Redewendungen*); and a 294-entry abbreviation list from the Wikipedia list of German abbreviations. The smaller closed lists (negation words, Konjunktiv forms, reader-addressing pronouns, W-question words, and cross-reference phrases) are enumerated with the rules that use them below.

Word level.

`Use Everyday Words` fraction of alphabetic tokens in a $\sim$4,600-word everyday-German list.

`Use Verbs Over Nouns` verb share of content words, $\text{VERB}/(\text{VERB} + \text{NOUN})$ (verbal vs. nominal style).

`Use Short Words` mean word length penalised against a corpus mean $(\max(0,\ 1 - \overline{\text{len}}/13.89))$.

`Avoid Abbreviations` $1-$ proportion of abbreviation tokens.

`Avoid Idioms` $1-$ share of tokens in a $\sim$2,700-entry idiom list.

`Use Positive Language` fraction of sentences with no negation word.

`Avoid Subjunctive Mood` fraction of sentences with no Konjunktiv form (*hätte, könnte, müsste, sollte, wäre, würde*).

`Use Active Voice` ratio of active to passive sentences (auxiliary + dependency heuristic).

Sentence level.

`Keep Sentences Short` 1 if mean sentence length $\leq$ 15.39 words, else a length penalty.

`Avoid Sentence Fragments` proportion of verb-less sentence fragments (the calibration absorbs the sign).

`Avoid Subordinate Clauses` fraction of sentences with no subordinate clause (`mark` dependency / subordinating conjunctions).

Text level.

`Address the Reader Directly` density of reader-addressing pronouns (*du, dir, dich, Sie, Ihnen, euch*).

`Minimize Questions` $1-$ fraction of W-question sentences.

`Avoid Cross-References` $1-$ fraction of sentences with a cross-reference phrase (*siehe, vgl., auf Seite*, ...).

Numbers and symbols (each $1-$ matches/tokens).

`Avoid Roman Numerals` Roman-numeral tokens.

`Avoid Historical Dates` years 1000–1899.

`Simplify Precise Numbers` grouped/decimal numbers (e.g. `1.000,45`).

`Use Digits for Numbers` spelled-out number words (simplified language prefers digits).

`Use Unambiguous Dates` ambiguous numeric dates (`DD.MM.YY`).

`Avoid Unnecessary Symbols` share of forbidden non-alphanumeric characters (allowed set `. , ! ? - –`, quotes, `EUR`).

B.2 Scope and assumptions

KLAR is defined for texts of at least 100 words; below that, readability statistics are unstable and the reliability weight degenerates. All evaluations use this $\geq$ 100-word (MW100) scope. All 20 rules carry equal weight, adjusted only for reliability. None of the rules, weights, or thresholds is fit to evaluation data, which is what keeps KLAR reference-free. The 20 rules cover the part of the *Leichte Sprache* rulebooks a deterministic check can measure; guidelines that need human judgment or document layout fall outside this, so KLAR operationalizes a subset of the rulebook rather than implementing it in full.

B.3 Readability indices (reported separately)

Computed on $\geq$ 100-word texts; never part of KLAR.

$$\text{FRE-DE} = 180 - \frac{\text{words}}{\text{sentences}} - 58.5 \cdot \frac{\text{syllables}}{\text{words}}$$

$$\text{WSTF}_4 = 0.2744\, MS + 0.2656\, \overline{\text{wps}} - 1.693$$

$$\text{LIX} = \overline{\text{wps}} + 100 \cdot \frac{\text{words}_{\geq 6\ \text{chars}}}{\text{words}}$$

where $\overline{\text{wps}}$ is mean words per sentence and MS is the *percentage* of words with ≥ 3 syllables (syllabification via PyHyphen 4.0.4 with the `de_DE` dictionary). The indices are reported only as independent points of comparison, never inside KLAR.

C Validation Detail Tables

Detail behind Section 5: per-corpus discrimination (Table 4), KLAR against the classical readability indices (Table 5), the normalization tradeoff decomposed step by step (Table 3), and the most diagnostic rules on Toborek (Table 6).

D Per-Rule Overlap with Readability Indices

Several of KLAR's rules are length-based, so the per-rule decomposition (Section 5.4) could in principle just re-encode the readability indices. It does not. A rule and an index built on the same surface feature must track each other, so agreement between them is overlap, not corroboration, and only one rule shows that overlap. `Use Short Words` correlates with the FRE, LIX, and WSF on every corpus ($\rho = 0.79$–0.94), always the top-correlated rule, while every other rule trails far behind. Table 7 gives, per corpus and index, the Spearman ρ of `Use Short Words` and, in parentheses, the next-highest rule. The overlap is a single shared length feature, so the rules as a set measure what the indices do not.

E Metric Comparison Details

Full numbers behind Figure 2 (Section 5).

Profiler re-implementation. The authors of Klöser et al. (2026) release no code, so we reconstruct the Simplification Profiler from the paper text and lock every open choice before scoring, so none can be tuned to the result. The rule-conformance axis (SIM) and grammar axis (LNG) are prose reconstructions: the paper's rule tables give no rule

identifiers, no LanguageTool version, and no subclause definition, so we implement SIM as deterministic spaCy-German detectors over those rules (long sentence > 15 words, long word > 3 syllables) and LNG as LanguageTool (de-DE, v6.8). The readability axis (FBR) leaves its slope undisclosed; we correlate on the raw axis, which is invariant under the monotone squashing the slope controls, so it cannot move any Spearman ρ. The meaning axes substitute our own model pins (the same NLI model and sentence embedder used elsewhere in this paper) because the paper names none, and our coverage axis is a deletion rate where theirs is a coverage score; this definitional difference is part of why the meaning axes diverge (Section 5.3). All five system outputs are scored over the same 997 MW100 documents.

Profiler convergence (Section 5.3). KLAR converges with the Profiler's independent readability axis, built from a different German formula ($\rho = -0.70$ against our FRE, negative because the Profiler scores complexity). Its rule-conformance axis also tracks KLAR ($\rho = 0.65$), but both encode the LanguageTool *Leichte Sprache* rulebook, so that agreement is shared construction, not independent support. The meaning and grammar axes, which each metric defines differently, diverge ($|\rho| < 0.3$).

F System Demonstration Tables

Exact values behind Figure 4 (Section 7). In the figure each axis is normalized so the best system reaches the outer ring; higher is better, and *Meaning* is shown as inverse content-deletion rate. The reference-based axes (SARI, BLEU, BERTScore-P) use the EASSE-DE pool ($n = 997$); FRE and Meaning use the generation set ($n = 514$). Each specialist falls out of the top band on at least one axis: ELGEPA on FRE, the *erlesen* models and German4all on BLEU.

G Human Evaluation Detail

Numbers behind the human preference result (Section 7) and the human-evaluation limitation (Section 9). Each item is one original document simplified two ways, once by Klarbot and once by one of the four fine-tuned specialists. Three linguistically trained raters (one PhD, two master's-level in phonetics and linguistics) saw the two simplifications and judged which better preserved meaning and which was more faithful.

We did not sample document pairs at random. Every pair already carried a verdict from the LLM-judge panel, and we oversampled the pairs the panel was split on, the cases where Klarbot and the specialist were closest and hardest to tell apart, so the sample spans the panel's full range of difficulty instead of only easy, lopsided pairs. Every agreement and preference figure below is therefore a conservative reading taken where judgment is hardest, not an average over easy and hard pairs alike. The primary numbers use three raters (181 pairwise and 158 absolute labels, 20 items co-rated by more than one rater); a fourth rater who labeled in a single rushed session is dropped, with the full four-rater run kept as a sensitivity.

H Register Placement Detail

Step-by-step separation behind the register ordering of Section 6. Each row is the change in KLAR across one register step, with a 95% cluster-bootstrap CI (cluster = sub-corpus); a step *separates* when its CI excludes zero. The human steps pool apa_lha, DEplain-web, and Toborek; the German4all steps are paired within document ($n = 514$, 17 sub-corpora). The ES→LS step is $+0.024$ on both, but only the larger, paired machine pool resolves it from zero.

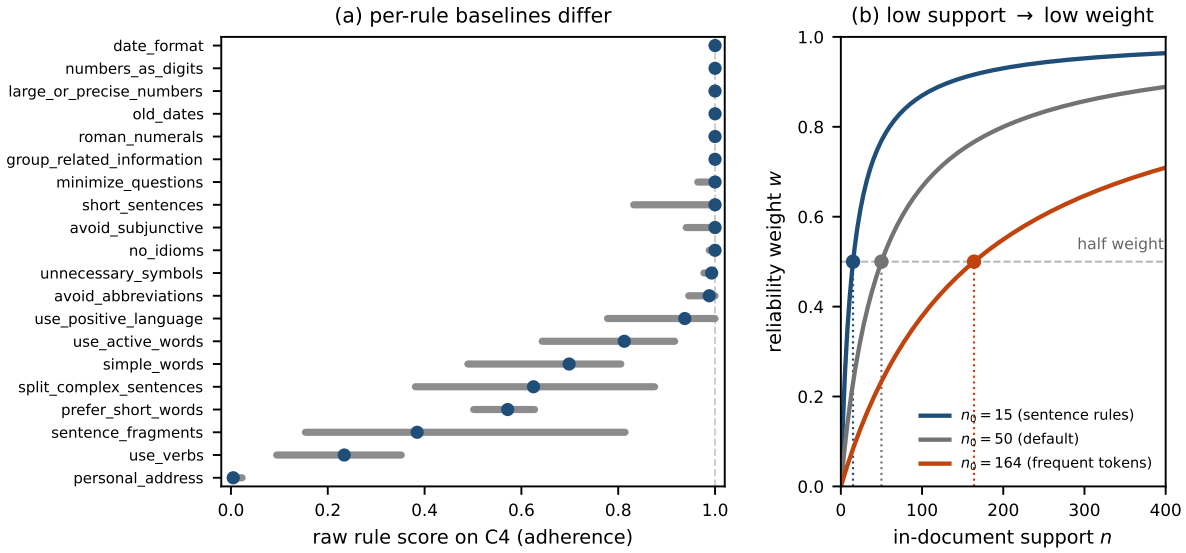


Figure 5: Why KLAR calibrates and weights (Section 4). **(a)** Each rule measures a different feature and has its own typical German level, estimated on C4. Some rules score near 1 on almost any text, others vary widely. A raw average would mix these incomparable scales, so we z-score each rule against its own C4 baseline and map every rule onto one deviation-from-typical scale. **(b)** The reliability weight $w = n/(n + n_0)$ rises with a rule's in-document support n. Each rule reaches half weight (dots) once n hits its n_0, which differs per rule (here 15, 50, 164); below that the rule is discounted, above it approaches full weight.

Corpus	n	Register / levels	Contrast	Domain	Source
APA-LHA (Spring et al., 2021)	208 / 1344	ES (B1, A2)	high	news	human
Toborek (Toborek et al., 2023)	369	ES + LS	high	encyclopedic	human
DEplain-web (Stodden et al., 2023)	172	ES	low	web	human
DEplain-APA (Stodden et al., 2023)	483	ES	low (hardest)	news	human
SimpEvalDE (Korobeynikova et al., 2026)	360 / 60 src	—	—	—	machine
ELG-CEFR-DE (Breuker, 2023; Imperial et al., 2025)	509	A1–C2	—	—	human
German4all (Anschütz et al., 2025)	514	ES + LS	—	—	machine

Table 2: Datasets used in this paper. *Contrast* is how far the simplified target moves from its original; *Source* is whether the simplifications are human-written or machine-generated. ES = *Einfache Sprache*, LS = *Leichte Sprache*; A1–C2 are CEFR reader levels, not registers. SimpEvalDE adds expert *simplicity* and *meaning* ratings on machine outputs.

Composite	Normalised	Weighted	Toborek	DEplain-web
Raw	—	—	0.986	0.698
Normalised	✓	—	0.966	0.709
KLAR	✓	✓	0.895	0.712

Table 3: The normalization tradeoff (Section 5.2), discrimination AUC on the two Toborek/DEplain-web corpora at MW100 ($n = 369$ and 172). Going from the raw combined score to KLAR, normalization costs $\Delta = -0.020$ on Toborek and reliability weighting a further $\Delta = -0.071$ (total $\Delta = -0.091$, 95% CI $[-0.203, -0.031]$; each step significant under cluster bootstrap). On low-contrast DEplain-web the three combined scores are equivalent. The cost is confined to the high-contrast case, and only the normalized scale lets each rule read as a deviation from typical German.

Corpus	n	AUC	% simpler
APA-LHA	208	0.946	99.5
Toborek	369	0.895	89.7
DEplain-web	172	0.712	83.0
DEplain-APA	483	0.652	73.7

Table 4: Discrimination of simplified vs. original text (MW100). AUC is the rank-separation of the two; % simpler is the share of pairs whose simplified text scores above its original. All Wilcoxon $p < 10^{-17}$.

Corpus	n	Fl.	LIX	WSF
Toborek	365	0.58	0.64	0.63
DEplain-web	172	0.17	0.16	0.15
DEplain-APA	483	0.21	0.15	0.16

Table 5: Convergence of KLAR with the classical indices: Spearman ρ of KLAR against the German Flesch Reading Ease (FRE), LIX, and WSF on three corpora. Convergence stays modest throughout, so KLAR and the formulas measure related but not identical things. n counts texts with computable indices ($\geq$ 100 alphabetic words), so it dips slightly below Table 4. Discrimination AUC for these indices is in Table 8.

Rule	mean Δ
`Keep Sentences Short`	$+0.56$
`Use Short Words`	$+0.43$
`Address the Reader Directly`	$+0.28$
`Use Positive Language`	$+0.26$
`Avoid Subjunctive Mood`	$+0.17$
`Minimize Questions`	-0.17

Table 6: The six most diagnostic rules on Toborek (MW100, $n = 369$); all BH-FDR $< 10^{-20}$. Positive Δ (simplified $-$ original) means the simplified text adheres more. Sentence splitting and short words dominate; question avoidance runs backwards.

Corpus	FRE	LIX	WSF
Toborek	0.94 (0.68)	0.87 (0.66)	0.92 (0.69)
DEplain-web	0.91 (0.64)	0.89 (0.68)	0.87 (0.63)
DEplain-APA	0.84 (0.19)	0.79 (-0.08)	0.81 (0.41)

Table 7: Per-rule Spearman ρ with each readability index. Each cell: `Use Short Words` (rank 1 everywhere, all FDR-significant) and, in parentheses, the next-highest rule. The shared word-length feature tracks the indices closely while every other rule trails sharply, so KLAR's convergence with the formulas is one feature, not redundancy.

			Discrimination AUC				Human
Metric	ref-free	explainable	APA-LHA	Toborek	DEplain-web	DEplain-APA	ρ
KLAR	yes	yes	0.946	0.895	**0.712**	0.652	0.31
FRE	yes	no	0.940	0.958	0.623	0.821	0.04
LIX	yes	no	0.930	0.969	0.608	0.809	0.23
WSF	yes	no	0.954	**0.982**	0.638	**0.846**	0.18
Profiler	yes	yes	0.880	0.967	0.698	0.741	0.29
DEtect	no	no		n/a (reference-based)			**0.35**

Table 8: Metric comparison behind Figure 2. Discrimination AUC (simplified vs. original) per corpus (APA-LHA, Toborek, DEplain-web, DEplain-APA news) and human-simplicity ρ on SimpEvalDE. Bold = column best. KLAR is the only reference-free, explainable metric and leads no quantitative column outright: the length formulas discriminate better where simplification is length reduction, and DEtect, which needs references, has the top ρ. Caveats: index AUC on APA-LHA is $n = 128$ (the indices' $\geq$ 100-alpha-word floor); the human-ρ texts are single sentences ($<$ 100 words), where the formulas are out of scope, so those cells are comparability points, not validity claims; Profiler shares KLAR's LanguageTool rulebook; DEtect is reference-based, so its discrimination cell is undefined.

System	SARI	BLEU	BERTScore-P
Klarbot	49.56	12.65	**0.363**
ELGEPA-8b	**52.65**	**16.86**	0.314
erlesen-leo-7b	45.16	6.52	0.305
erlesen-leo-13b	44.09	4.54	0.261
German4all	39.79	0.24	0.322

Table 9: Reference-based comparison (EASSE-DE, $n = 997$: Toborek, DEplain-web, DEplain-APA). Bold = column-leading point estimate; see Section 7 for significance, where Klarbot–ELGEPA is a tie on both SARI and BERTScore-P.

		Content-deletion ↓	
System	FRE	mDeBERTa	XLM-R
Klarbot	85.2	**0.644**	0.621
ELGEPA-8b	63.2	0.688	0.676
erlesen-leo-7b	83.9	0.649	**0.607**
erlesen-leo-13b	**87.7**	0.708	0.678
German4all	67.7	0.799	0.761

Table 10: The two reference-free axes of Figure 4, exact values ($n = 514$ generation set). FRE is the independent readability score (higher is simpler); content-deletion is the fraction of original sentences left unsupported by the output, scored by two German NLI models (lower is better), the *Meaning* axis shown inverted in the figure. Bold = column-leading point estimate. Klarbot has the lowest deletion rate under mDeBERTa and ties erlesen-leo-7b under XLM-R; the two models agree per document at Pearson $r = 0.64$ to 0.94 across systems. On FRE Klarbot is statistically equivalent to both *erlesen* models (TOST, ± 5; Section 7).

Axis	Klarbot	tie	specialist
Meaning	99	68	14
Faithfulness	69	99	13

Table 11: Human pairwise preference ($n = 181$), pooled over the four fine-tuned specialists (42 pairs each). Even on this contested-enriched sample the raters lean toward Klarbot and almost never a specialist.

Axis	Human ceiling	Panel↔human [95% CI]	Co-rated
Meaning	0.20	0.23 [0.12, 0.33]	0.36
Faithfulness	0.28	0.11 [0.02, 0.20]	0.03

Table 12: Agreement (Cohen's κ). *Human ceiling* is the pooled human–human κ; *Panel↔human* is the pooled bootstrap estimate ($n = 181$); *Co-rated* compares the panel to humans on the same 20 items. On meaning the panel (0.36) meets the human ceiling (0.20); on faithfulness (0.03) it falls short (0.28).

Human rating	Composite ρ	Composite β/SD	FRE β/SD
Standalone quality	0.21	0.19 [0.09, 0.29]	0.17
Meaning preserved	0.11	0.06 [−0.02, 0.14]	0.06

Table 13: The combined-score *level* against the raters' absolute ordinal ratings. ρ is the rank correlation ($n = 316$); β/SD is a per-standard-deviation OLS slope controlling for rater (fixed effect), with a cluster-robust CI on doc. On quality the combined score matches Flesch Reading Ease; on meaning both include zero, controlling for rater or not, so the score reads simplicity, not meaning.

Step	ΔKLAR	95% CI	separates
Human registers			
StS → ES	+0.115	[0.064, 0.150]	yes
ES → LS	+0.024	[−0.007, 0.070]	no
German4all (machine, $n = 514$)			
original → ES	+0.066	[0.050, 0.078]	yes
ES → LS	+0.024	[0.010, 0.038]	yes

Table 14: Register-step separation (Section 6). German4all arm means: original 0.504, ES 0.569, LS 0.593. The ES→LS step is the one close call on human text, where its CI crosses zero; it separates on the larger paired machine pool.